

IBM

Pill Tracker Lab

Quick Reference Guide

Division: IBM Client Engineering

Solution: Healthcare Bob Demo

Version: 1.0

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Overview

This lab demonstrates how IBM Bob accelerates full-stack application development by transforming design mockups into production-ready code. You'll build a complete medication tracking system with Drupal backend, MySQL database, and responsive frontend.

Learning Objectives:

- Convert design mockups to functional HTML/CSS
- Iterate on designs with AI assistance
- Generate full-stack applications from specifications
- Validate and deploy production-ready code

Prerequisites

Required Tools

- **IBM Bob IDE** - Installed and configured in VS Code
- **Web Browser** - For HTML testing and application access
- **Docker Desktop** - For running the Drupal stack
- **Design Files** - `initial_pill_tracker.png` and `edited_pill_tracker.png`

Step 1: Install Docker

macOS Installation

1. Download Docker Desktop from <https://www.docker.com/products/docker-desktop>
2. Open the downloaded `.dmg` file
3. Drag Docker to Applications folder
4. Launch Docker Desktop from Applications
5. Follow the setup wizard and grant necessary permissions

Windows Installation

1. Download Docker Desktop from <https://www.docker.com/products/docker-desktop>
2. Run the installer
3. Enable WSL 2 if prompted
4. Restart your computer
5. Launch Docker Desktop

Linux Installation

```
# Ubuntu/Debian
sudo apt-get update
sudo apt-get install docker.io docker-compose
sudo systemctl start docker
sudo systemctl enable docker

# Add your user to docker group
sudo usermod -aG docker $USER
# Log out and back in for group changes to take effect
```

Verify Installation

```
docker --version
docker-compose --version
```

✓ **Expected Output:** Docker version 20.x or higher and Docker Compose version 1.29 or higher.

⚠ **Important:** Ensure Docker Desktop is running before proceeding with the lab. You should see the Docker icon in your system tray/menu bar.

Step 2: Environment Setup

Create Project Directory

```
mkdir pill-tracker-lab  
cd pill-tracker-lab
```

Prepare Design Files

Place the provided PNG design files in your project directory:

- [initial_pill_tracker.png](#) - Original design mockup
- [edited_pill_tracker.png](#) - Updated design with refinements

Verify Bob Connection

1. Open VS Code in your project directory
2. Open Bob chat panel: `Cmd/Ctrl + Shift + P` → "Bob: Open Chat"
3. Test connection with: *"Hello, are you ready to help with the Pill Tracker lab?"*

✓ **Checkpoint:** Bob should respond confirming readiness to assist with the lab.

Step 3: Design to HTML Conversion

PRIMARY PROMPT

Create a functional HTML file using the provided PNG file titled "initial_pill_tracker.png". Keep everything looking as close to the original as possible--including headers, colors, buttons, and sections--and use placeholders for anything that can't be accessed.

Follow-up Prompts

UNDERSTANDING STRUCTURE

Explain the structure of pill_tracker.html. What are the main sections and how is the styling organized?

CSS ANALYSIS

What CSS properties were used to create the gradient background? Can you show me the exact color values?

Actions

1. Provide `initial_pill_tracker.png` to Bob with the primary prompt
2. Review the generated `pill_tracker.html` file
3. Open in browser: `open pill_tracker.html`

✓ **Expected Result:** Pixel-perfect HTML recreation of the design mockup with gradient background, medication cards, and interactive elements.

Step 4: Design Iteration

PRIMARY PROMPT

Edit the HTML with the changes made in the new PNG file titled "edited_pill_tracker.png". Make sure to keep everything looking as close to the original as possible.

Follow-up Prompts

CHANGE ANALYSIS

Show me the exact changes you made to pill_tracker.html. What was modified?

DOCUMENTATION

Create a changelog entry documenting the changes made to pill_tracker.html in this iteration.

Actions

1. Provide `edited_pill_tracker.png` to Bob
2. Review the diff and approve changes
3. Refresh browser to see updates

Key Learning: Bob can intelligently compare designs and apply only the necessary changes while preserving existing structure and functionality.

Step 5: Full-Stack Application Generation

PRIMARY PROMPT

Based on this HTML design, create a drupal stack for the backend with a dummy mysql database & a web front end.

What Bob Creates

Infrastructure

- `docker-compose.yml` - Multi-container orchestration
- `database/init.sql` - Schema and dummy data
- `start.sh` / `stop.sh` - Management scripts

Backend (Drupal Module)

- Complete module structure with metadata
- Database service class with CRUD operations
- Dashboard and API controllers
- 6 RESTful API endpoints

Frontend

- Twig template matching original design
- CSS styling (279 lines)
- JavaScript with AJAX integration

Follow-up Prompts

DATABASE SCHEMA

Show me the SQL schema for the pill tracker database file. What tables were created and what are their primary keys?

CONTROLLER ANALYSIS

What are the main functions in pill tracker controller php file and what does each one do?

⚠ **Note:** This step generates 20+ files and ~2,000 lines of code. Review the file structure before proceeding.

Step 6: Launch Application

OPTIONAL PROMPT (IF SCRIPTS NOT GENERATED)

Create start.sh and stop.sh scripts to manage the Docker containers for the Pill Tracker application.

Note: If Bob didn't automatically create `start.sh` and `stop.sh` in Step 3, use the prompt above to generate them before proceeding.

Start the Stack

```
./start.sh
```

Wait 2-3 minutes for containers to initialize. The script will:

- Pull Docker images (Drupal, MySQL, phpMyAdmin)
- Create network and volumes
- Initialize database with schema and dummy data
- Start all services

⚠ **Troubleshooting:** If you encounter errors like "container state improper" or MySQL fails to start:

1. Clean up everything: `docker-compose down -v`
2. Remove orphaned containers: `docker ps -a | grep pill_tracker | awk '{print $1}' | xargs docker rm -f`
3. Try starting again: `./start.sh`
4. If issues persist, check port 3306 availability: `lsof -i :3306`

Verify Database

```
docker exec pill_tracker_mysql mysql -u root -prootpassword \  
-e "USE pill_tracker_db; SELECT * FROM patients;"
```

✓ **Expected Output:** Three patients (Sarah Johnson, Michael Chen, Emily Rodriguez) with complete medication data.

Access Points

- **Drupal:** <http://localhost:8080>
- **phpMyAdmin:** <http://localhost:8081>
- **Dashboard:** <http://localhost:8080/pill-tracker> (after setup)

Step 7: Drupal Installation

1. Navigate to <http://localhost:8080>
2. Follow installation wizard
3. **Critical:** In database configuration, click "Advanced options"
4. Use these credentials:
 - Database: [pill_tracker_db](#)
 - Username: [drupal](#) (no spaces before or after)
 - Password: [drupal123](#)
 - Host: [mysql](#) (not localhost!)
5. Configure site settings:
 - Site name: [Pill Tracker](#)
 - Site email: [admin@pilltracker.local](#) (any format works for local development)
 - Admin username: [admin](#) (you'll use this to log in)
 - Admin password: [admin123](#) (choose something memorable)
 - Admin email: [admin@pilltracker.local](#)
6. Click "Save and continue" to complete installation

 **Note:** Site and admin email addresses are used for automated emails. Since this is a local development environment, any email format works (e.g., [admin@pilltracker.local](#)). Write down your admin credentials - you'll need them to enable the Pill Tracker module.

 **Important:** Use [mysql](#) as the database host, not [localhost](#) . This is the Docker service name.

Step 8: Enable Module

1. Navigate to <http://localhost:8080/admin/modules>
2. Find "Pill Tracker" under Custom section
3. Enable the module
4. Clear cache: <http://localhost:8080/admin/config/development/performance>
5. Visit dashboard: <http://localhost:8080/pill-tracker>

✓ **Success:** You should see the fully functional Pill Tracker dashboard with patient data, medication cards, and working buttons.

Troubleshooting: Buttons Not Working

⚠ **Issue:** If buttons don't respond or you see JavaScript errors in the browser console (F12), this is likely a browser cache issue with Drupal's aggregated JavaScript files.

Solution 1: Hard Refresh Browser (Recommended)

Force your browser to reload all cached files:

- **Windows/Linux:** Press `Ctrl + Shift + R` or `Ctrl + F5`
- **Mac:** Press `Cmd + Shift + R` or `Cmd + Option + R`

Solution 2: Use Incognito/Private Window

Open the dashboard in a private browsing window to bypass all caches:

- **Chrome:** `Ctrl/Cmd + Shift + N`
- **Firefox:** `Ctrl/Cmd + Shift + P`
- **Safari:** `Cmd + Shift + N`

Solution 3: Disable Cache in DevTools

1. Open Developer Tools (F12)
2. Go to Network tab
3. Check "Disable cache" checkbox
4. Keep DevTools open and refresh the page

Solution 4: Disable Drupal Aggregation (For Development)

1. Navigate to <http://localhost:8080/admin/config/development/performance>
2. Uncheck "Aggregate JavaScript files"
3. Uncheck "Aggregate CSS files"
4. Click "Clear all caches"
5. Click "Save configuration"
6. Hard refresh your browser

💡 **Why This Happens:** Drupal combines and minifies JavaScript files for performance. When code changes, the aggregated files need to be regenerated. Browser caching can serve old versions even after clearing Drupal's cache. Hard refreshing forces the browser to fetch new files.

Step 9: Validation

PRIMARY PROMPT

Please validate the "pill_tracker/*.php" drupal stack for health and functionality.

BEST PRACTICES REVIEW

Review the Pill Tracker module for Drupal best practices. Are there any anti-patterns or areas for improvement?

Test Functionality

- Switch between patients
- Mark medications as taken
- Verify status updates (blue = taken, yellow = due, red = overdue)
- Check browser console for errors
- Inspect network requests in DevTools

Common Bob Interaction Patterns

Request Explanations

Explain the structure of [filename]. What are the main sections?

Request Changes

Update [filename] to change [specific detail].

Troubleshooting

The [action] failed with error: [paste error]. Can you fix it?

Verification

Compare [file1] with [file2]. Are they consistent?

Key Takeaways

- **Visual Understanding:** Bob analyzes design mockups and generates pixel-perfect code
- **Iterative Development:** Easy refinement of designs and features
- **Full-Stack Generation:** Complete applications from infrastructure to frontend
- **Production Quality:** Clean, documented, best-practice code
- **Time Savings:** 10-12 hours of work completed in under 1 hour

Business Impact: This workflow demonstrates how IBM Bob accelerates development cycles, reduces costs, and enables rapid prototyping for client engagements.

Lab Cleanup

Stop the Application

When you're finished with the lab, stop all Docker containers:

```
./stop.sh
```

This script will:

- Stop all running containers (Drupal, MySQL, phpMyAdmin)
- Remove containers
- Preserve data volumes for future use

Complete Cleanup (Optional)

To completely remove all containers, volumes, and data:

```
docker-compose down -v
```

⚠ **Warning:** The `-v` flag deletes all data volumes, including the database. Only use this if you want to start completely fresh or are done with the lab permanently.

Restart After Cleanup

To restart the application after stopping:

```
./start.sh
```

💡 **Note:** If you used `docker-compose down -v`, you'll need to reinstall Drupal (Step 5) and re-enable the module (Step 6) since all data was deleted.



